习题15.1 p256 1-8任选两题，9,10，11-21任选两题
9题和10题的进一步思考:验证结论成立实属不难，但如何想到这样的函数的?
习题15.2 p260 1-11任选三题
习题15.3 p268 1,2,5,6,8
进一步思考:这些题目大都是“验证”一个函数满足一个pde，如果把这个问题倒过来看(求解pde)，又是如何“猜”出这些可能的函数形式呢?(物理?几何?甚至连蒙带猜?)
习题15.4p272 任选三道题
习题15.5 p284 任选五道题
习题15.6 p299 任选三道题
习题15.7 p305 1-10任选三题，11-22任选三题
习题15.8 p312 任选三题.

15.1

15.1.1

Suppose $f:\mathbb{R}^n\to\mathbb{R}^m$, prove that if u,v are unit vectors and $u=-v$ then $\frac{\partial f}{\partial u}=-\frac{\partial f}{\partial v}$.
Proof: By definition, $\frac{\partial f}{\partial u}=\frac{d}{dt}f(x+tu)=-\frac{d}{dt}f(x-tu)=-\frac{\partial f}{\partial v}$.

15.1.5

Suppose the partial derivatives of f exist and are bounded, prove that f is continuous.
Proof: For $\mathbf{x}$ and $\mathbf{h}\to 0$, suppose $\mathbf{h}_i=\mathbf{h}^1+\cdots+\mathbf{h}^i$ then $f(\mathbf{x}+\mathbf{h})-f(\mathbf{x})$
$=\sum_{i=1}^n f(\mathbf{x}+\mathbf{h_i})-f(\mathbf{x}+\mathbf{h}_{i-1})=\sum_{i=1}^n h_i\frac{\partial f}{\partial x^i}(\mathbf{x}+\xi_i)$ tends to zero.

15.1.9

Consider $u(x,y)=e^x\cos y$ and $v(x,y)=e^x\sin y$, prove that u,v are harmonic.
Proof: Clearly $\exp$ is holomorphic, so $u=\mathrm{Re}(\exp)$ is harmonic. By symmetry v is harmonic.

15.1.10

Let $u:\mathbb{R}^3\to\mathbb{R},\mathbf{x}\mapsto 1/|\mathbf{x}|$, prove that u is harmonic.
Proof: For a radical function, $\Delta u=\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}u\right)$, so clearly $\Delta u=0$ in $\mathbb{R}^3\backslash\{0\}$. Generally, for $n\geqslant 2$, $u:x\mapsto|\mathbf{x}|^{2-n}$ is harmonic, and for $n=2$, $u:x\mapsto\log|\mathbf{x}|$ is harmonic.

15.1.11

Suppose $f:\mathbb{R}^2\to\mathbb{R}$ is continuous in both coordinates, and $f(\cdot,y)$ is monotone for all y, prove that f is continuous.
Proof: It suffices to verify $f^{-1}(a,b)$ is open. For any $(x_0,y_0)\in f^{-1}((a,b))$, take $x_1<x_0<x_2$ such that $a<f(x_1,y_0)<f(x_0,y_0)<f(x_2,y_0)<b$, then since $f(x,\cdot)$ is continuous, there exists $\varepsilon>0$ such that $f(x_1,y)>a$ and $f(x_2,y)<b$ for all. $|y-y_0|<\varepsilon$. Hence $f^{-1}((a,b))$ contains the open neighborhood $(x_1,x_2)\times(y_0-\varepsilon,y_0+\varepsilon)$, so $f^{-1}((a,b))$ is open.

15.1.20

Suppose $\{\ell_1, \cdots, \ell_n\}$ are linearly independent unit vectors, D open connected, and $f : D \subset \mathbb{R}^n \to \mathbb{R}$ is differentiable and satisfy $\frac{\partial f}{\partial \ell_i} = 0$ for all i, then f is constant.
Proof: $\{\ell_1, \cdots, \ell_n\}$ form a basis of the tangent space $\mathbb{R}^n$, so $\frac{\partial f}{\partial \ell_i} = 0$ implies $df \equiv 0$, hence f is constant.

15.2

15.2.2

Prove that $f(x, y) = \sqrt{|xy|}$ is not differentiable at 0.
Proof: Otherwise $\sqrt{|xy|} = df(xe_1 + ye_2) + o(|x| + |y|)$, so $\sqrt{|xy|} = xa_1 + ya_2 + o(|x| + |y|)$ where $a = df(e_1)$ and $b = df(e_2)$. Clearly this is impossible, since fix $y = 0$ we have $a_1 = 0$ and likewise $a_2 = 0$, but let $x = y$ we have $a_1 + a_2 = 1$.

15.2.7

Suppose $f : [a, b] \to \mathbb{R}^n$ and for any $t \in [a, b]$, $|f(t)| = C$. Prove that $\langle f', f\rangle = 0$.
Proof: Differentiating $f \cdot f \equiv C$ gives $f' \cdot f \equiv 0$.

15.2.10

Suppose $f : \mathbb{R}^n \to \mathbb{R}^m$ is linear, determine Jf.
Solution: Clearly $df = f$ since $f(x + h) = f(x) + f(h)$ for any x, h. Hence Jf is the matrix of f under standard basis.

15.3

15.3.1

Suppose $f(x, y, z) = F(u, v, w)$, where $x^2 = vw$, $y^2 = wu$, $z^2 = uv$, prove that

$$x\frac{\partial f}{\partial x} + y\frac{\partial f}{\partial y} + z\frac{\partial f}{\partial z} = u\frac{\partial F}{\partial u} + v\frac{\partial F}{\partial v} + w\frac{\partial F}{\partial w}.$$

Proof: Consider $g(t) = f(t(x, y, z)) = F(t(u, v, w))$ (since (x, y, z) and (u, v, w) are homogenous), then by the chain rule, $dg = (x, y, z) \cdot df = (u, v, w) \cdot dF$. Evaluating at $t = 1$ we obtain $\sum x\frac{\partial f}{\partial x} = \sum u\frac{\partial F}{\partial u}$.

15.3.2

Suppose $f : \mathbb{R}^3 \to \mathbb{R}$, suppose e_1, e_2, e_3 are orthonormal vectors in $\mathbb{R}^3$, prove that

$$\sum_{i=1}^{3}\left(\frac{\partial f}{\partial e_i}\right)^2 = \left(\frac{\partial f}{\partial x}\right)^2 + \left(\frac{\partial f}{\partial y}\right)^2 + \left(\frac{\partial f}{\partial z}\right)^2.$$

Proof: Both are $(df)^T \cdot (df)$, so they are equal since df is independent of choice of coordinates.

15.3.5

Prove that if $u : \mathbb{R}^2 \to \mathbb{R}$ is harmonic, then so is $v(x, y) = u(x^2 - y^2, 2xy)$ and $w(x, y) = u\left(\frac{x}{x^2+y^2}, \frac{y}{x^2+y^2}\right)$.
Proof: If $z = x + iy$, then $z^2 = x^2 - y^2 + 2xyi$, hence u is harmonic implies $u = \mathrm{Re} f$ for holomorphic f, so $v = \mathrm{Re}(f \circ z^2)$ is harmonic. Analogously, $\frac{1}{\bar{z}} = \frac{x}{x^2+y^2} + i\frac{y}{x^2+y^2}$ is conformal, so $w = \mathrm{Re}\left(f \circ \frac{1}{\bar{z}}\right)$ is harmonic.
We can actually show that $w(\mathbf{x}) = |\mathbf{x}|^{n-2} u(\mathbf{x}/|\mathbf{x}|^2)$ is harmonic for any $u : \mathbb{R}^n \to \mathbb{R}$ harmonic.

15.3.6

Prove that $u(x, t) = \frac{1}{2a\sqrt{\pi t}} e^{-(x-b)^2/4a^2t}$ satisfy $\frac{\partial u}{\partial t} = a^2 \frac{\partial^2 u}{\partial x^2}$.
Proof: u is the heat kernel on the real line.
To obtain this solution, WLOG assume $a = 1, b = 0$, then $u(x, t) = \frac{1}{2\sqrt{\pi t}} e^{-x^2/4t}$. By Gaussian identity $\hat{u}(\xi, t) = e^{-t\xi^2}$ so $\widehat{\frac{\partial^2 u}{\partial x^2}} = \xi^2 \hat{u}(\xi, t) = -\xi^2 e^{-t\xi^2}$ and $\frac{\partial}{\partial t}\hat{u}(\xi, t) = -\xi^2 e^{-t\xi^2}$. Hence taking inverse Fourier transformations we have $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$.

15.3.8

Suppose $\varphi, \psi \in C^2(\mathbb{R})$, prove that $u(x, y) = \varphi\left(\frac{y}{x}\right) + y\psi\left(\frac{y}{x}\right)$ satisfy the PDE

$$x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = 0. \tag{$\star$}$$

Proof: It suffices to verify $u = \varphi\left(\frac{y}{x}\right)$ and $u = y\psi\left(\frac{y}{x}\right)$.
For $u(x, y) = \varphi\left(\frac{y}{x}\right)$, we have

$$\frac{\partial^2 u}{\partial x^2} = \varphi''\left(\frac{y}{x}\right)\frac{y^2}{x^4} + \varphi'\left(\frac{y}{x}\right)\frac{2y}{x^3}, \quad \frac{\partial^2 u}{\partial x \partial y} = -\varphi''\left(\frac{y}{x}\right)\frac{y}{x^3} - \varphi'\left(\frac{y}{x}\right)\frac{1}{x^2}$$

and $\frac{\partial^2 u}{\partial y^2} = \varphi''\left(\frac{y}{x}\right)\frac{1}{x^2}$. Hence u satisfy $(\star)$.
Analogously $u(x, y) = y\psi\left(\frac{y}{x}\right)$ satisfy $(\star)$.
To explain why, consider $D = x\frac{\partial}{\partial x} + y\frac{\partial}{\partial y}$, then $(\star)$ is just $D(D-1)u = 0$. By Euler's identity, $u(x, y) = \varphi\left(\frac{y}{x}\right)$ is homogenous so $Du = 0$, and $u(x, y) = y\psi\left(\frac{y}{x}\right)$ is homogenous of degree 1, so $Du = u$.

15.4

15.4.8

Suppose $f : [a, b] \to \mathbb{R}^n$ and $g : [a, b] \to \mathbb{R}$ are continuous, and both are differentiable on (a, b). If $|f'(t)| \leqslant g'(t) \forall t \in (a, b)$, prove that $|f(b) - f(a)| \leqslant g(b) - g(a)$.
Proof: If $f(b) = f(a)$ the result is trivial.
Let $e = \frac{f(b)-f(a)}{|f(b)-f(a)|}$, and consider $h : [a, b] \to \mathbb{R}, x \mapsto (f(x) - f(a)) \cdot e$. Then $h(a) = 0$ and $h(b) = |f(b)$

$- f(a)|$, $|h'(x)| = |f'(x) \cdot e| \leqslant |f'(x)| \leqslant g'(x)$. Hence by 1-dim version of finite increment theorem, $|f(b) - f(a)| = |h(b) - h(a)| \leqslant g(b) - g(a)$.

15.4.10

Suppose $f \in C^2(\mathbb{R}^n)$, $A \in \mathbb{R}^{n\times n}$ is orthonormal, and $F(x) = f(Ax)$. Prove that for $y = Ax$, we have

$$\sum_i \left(\frac{\partial f}{\partial y_i}\right)^2 = \sum_i \left(\frac{\partial F}{\partial x_i}\right)^2, \quad \sum_i \frac{\partial^2 f}{\partial y_i^2} = \sum_i \frac{\partial^2 F}{\partial x_i^2}.$$

Proof: $df_y = A \cdot dF_x$ so $(df_y)^T \cdot (df_y) = (dF_x^T A^T) \cdot (AdF_x) = (dF_x)^T (dF_x)$. Likewise $d(df_y) = A^T d(dF_x) A$ so $\Delta f = \mathrm{tr}(d(df_y)) = \Delta F$.

15.4.11

Suppose $f : \mathbb{R}^n \to \mathbb{R}^n$ is differentiable, and $\|df_x - id\| \leqslant k \forall x \in \mathbb{R}^n$ where $k \in (0,1)$. Prove that f is injective, and the preimage of bounded sets are bounded.
Proof: For any $x_1 \in \mathbb{R}^n$, $g(x) = -f(x) + f(x_1) + x$ satisfy $\|dg_x\| = \|df_x - id\| \leqslant k$, so by Banach fixed point theorem g has a unique fixed point in $\mathbb{R}^n$. Hence $f(x) = f(x_1)$ has a unique solution in $\mathbb{R}^n$, so f is injective.
Note that $|f(x) - f(0)| \geqslant (1 - k)|x - 0|$, so the preimage of bounded sets are bounded.

15.5

15.5.10

Suppose $f : D \to \mathbb{R}$ is differentiable where $D \subset \mathbb{R}^n$ is convex, prove that f is convex iff

$$(\nabla f(x) - \nabla f(y)) \cdot (x - y) \geqslant 0 \forall x, y \in D.$$

Proof: If f is convex, then $f(y) - f(x) \geqslant \nabla f(x) \cdot (y - x)$ and $f(x) - f(y) \geqslant \nabla f(y) \cdot (x - y)$, so $(\nabla f(x) - \nabla f(y)) \cdot (x - y) \geqslant 0$.
Conversely, for any $x, y \in D$, take ξ such that $f(y) - f(x) = \nabla f(\xi) \cdot (y - x)$, then $(\nabla f(\xi) - \nabla f(x))(x - y) \geqslant 0$ so $f(y) - f(x) \geqslant \nabla f(x) \cdot (y - x)$.

15.5.16

Determine the maximum&minimum of $f(x, y) = xy\sqrt{1 - \frac{x^2}{a^2} - \frac{y^2}{b^2}}$.
Solution: Note that $xy \leqslant \frac{ab}{2}\left(\frac{x^2}{a^2} + \frac{y^2}{b^2}\right)$, so $f(x, y) \leqslant \frac{ab}{2}\left(\frac{x^2}{a^2} + \frac{y^2}{b^2}\right)\sqrt{1 - \frac{x^2}{a^2} - \frac{y^2}{b^2}} \leqslant \frac{ab}{3\sqrt{3}}$ is the maximum value, and the minimum clearly $-\frac{ab}{3\sqrt{3}}$.

15.5.23

Suppose $D \subset \mathbb{R}^n$ is bounded open connected and has smooth boundary. If $f : \overline{D} \to \mathbb{R}$ is C^2 such that $\Delta f(x) > 0$, prove that f cannot reach its maximum in D.
Proof: $\Delta f > 0$ means f is a subharmonic function, which satisfy the mean-value property:

$$f(x) \leqslant \frac{1}{m(\partial B(x,r))} \int_{\partial B(x,r)} f \,\mathrm{d}m.$$

Hence if $f(x)$ reaches maximum M at $x_0 \in D$, then $f^{-1}(\{M\}) \cap D$ is both open and closed. Since D is connected, f is constant on D, a contradiction.

15.5.24

Suppose $D \subset \mathbb{R}^n$ is bounded closed connected and $u : D \to \mathbb{R}$ is C^2 in $\mathrm{int}(D)$ such that $\Delta u + cu = 0$ for some constant $c < 0$. Prove that u cannot reach its positive maximum in the interior of D, and if u is continuous on D, vanishes on ∂D, then $u \equiv 0$ on D.
Proof: At an interior positive maximum, $\Delta u < 0$, but the PDE gives $\Delta u = -cu > 0$, a contradiction.
If u is continuous on D and vanishes on ∂D, by Stoke's theorem,

$$\begin{aligned} 0 &= \int_{\partial D} u \cdot \nabla u \,\mathrm{d}m = \int_D \nabla (u \cdot \nabla u) \,\mathrm{d}m = \int_D u \cdot \Delta u \,\mathrm{d}m + \int_D |\nabla u|^2 \,\mathrm{d}m \\ &= (-c) \int_D |u|^2 \,\mathrm{d}m + \int_D |\nabla u|^2 \,\mathrm{d}m. \end{aligned}$$

Therefore $u \equiv 0$ on D.

15.6

15.6.5

Suppose $f : \mathbb{R}^n \to \mathbb{R}^n$ is C^1, and $|f(x) - f(y)| \geqslant |x - y| \forall x, y \in \mathbb{R}^n$. Prove that f is invertible and f^{-1} is also C^1.
Proof: Clearly $|df_x(h)| \geqslant |h|$ so df_x is non-singular and f has a local C^1 inverse everywhere. It suffices to show f is surjective, then we can patch up all local inverses to obtain a global inverse.
Since f is a local diffeomorphism, it is an open mapping. If $f(x_n) \to a$, then $\{f(x_n)\}$ is a Cauchy sequence so $\{x_n\}$ is a Cauchy sequence, hence $a = f(\lim_{n\to\infty} x_n)$ and $f(\mathbb{R}^n)$ is closed. Therefore $f(\mathbb{R}^n)$ is both open and closed, so f is surjective.

15.6.8

Given $f(x, y, z) = 0$, suppose $x = x(y, z), y = y(x, z), z = z(x, y)$ has continuous derivatives, prove that

$$\frac{\partial x}{\partial y} \frac{\partial y}{\partial z} \frac{\partial z}{\partial x} = -1.$$

Proof: Note that $\frac{\partial x}{\partial y} = -\frac{\partial f}{\partial y} / \frac{\partial f}{\partial x}$, so their product is $(-1)^3 = -1$.

15.6.12

Suppose $F:\mathbb{R}^2\to\mathbb{R}$ is C^2, and the curve $F(x,y)=0$ is the figure eight curve. What is the least number of critical points of F?
Solution: The answer is 3. The figure-eight curve is not locally Euclidean at the center, so the center must be a critical point, otherwise by the implicit function theorem $F^{-1}(0)$ is locally Euclidean. By Rolle's theorem there are two critical points inside each bounded component of $F^{-1}(\mathbb{R}\backslash\{0\})$.
The bound is sharp, given by $F(x,y)=(x^2+y^2)^2-x^2+y^2$.

15.7

15.7.13

For $a_1,\cdots,a_n\geqslant 0$, find the maximum and minimum value of $u=\sum a_ix_i$ on the sphere $\sum x_i^2=r^2$.
Solution: By Cauchy's inequality, the maximum is $r\sqrt{\sum a_i^2}$ and the minimum $-r\sqrt{\sum a_i^2}$.

15.7.18

Find the critical values of $f(x,y)=\sin x+\cos y+\cos(x-y)$ in $\left[0,\frac{\pi}{2}\right]^2$.
Solution: The only interior critical point is $\left(\frac{\pi}{3},\frac{\pi}{6}\right)$ and its value is $\frac{3\sqrt{3}}{2}$.

15.7.20

Find the critical values of $z=z(x,y)$ determined by the implicit function $2x^2+y^2+z^2+2xy-2x-2y-4z+4=0$.
Solution: $\frac{\partial z}{\partial x}=\frac{\partial z}{\partial y}=0$ iff $\frac{\partial F}{\partial x}=\frac{\partial F}{\partial y}=0$, i.e. $z=1,3$.

15.8

15.8.7

Prove that all tangent planes of $z=xe^{x/y}$ pass the origin.
Proof: Let $F(x,y,z)=xe^{x/y}-z$ then $\nabla F(x,y,z)=\left(e^{x/y}+\frac{x}{y}e^{x/y},-\frac{x^2}{y^2}e^{x/y},-1\right)$ so $(x,y,z)\cdot\nabla F(x,y,z)\equiv 0$. Therefore the tangent plane passes the origin.

15.8.13

Prove that all tangent spaces (planes) of $F(x-az,y-bz)=0$ is parallel to some line, where a,b are constants.
Proof: Clearly for $G(x,y,z)=F(x-az,y-bz)$, $(a,b,1)\perp\nabla G$ so every tangent plane is parallel to the line with direction $(a,b,1)$.

15.8.14

Prove that the tangent planes of $F\left(\frac{x-a}{z-c}, \frac{y-b}{z-c}\right)=0$ pass a common point, where a, b, c are constants.
Proof: Analogously, $G(x, y, z)=F\left(\frac{x-a}{z-c}, \frac{y-b}{z-c}\right)$ satisfy $(\mathbf{x}-(a, b, c)) \cdot \nabla G \equiv 0$ so they pass the point (a, b, c).